BRANCHES OF A ONE-DIMENSIONAL NOETHERIAN LOCAL RING AGREE WITH ITS COMPLETION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The number of branches of a Noetherian local ring is at most that of its completion. invoked at proof lines 32216-32217. One direction of the equality.
- Equality of branch counts is equivalent to every minimal prime of the henselization staying prime after completion, and in dimension one every formal minimal prime comes from the henselization. invoked at proof lines 32217-32219. The other direction of the equality.
- The completion of a one-dimensional local ring is one-dimensional. invoked at proof lines 32220-32221. Supplies the dimension hypothesis for the formal-branch lifting.
- Strict henselization does not change the dimension, and the completion of the strict henselization agrees with that of the completion's strict henselization. invoked at proof lines 32223-32227. Transfers the branch result to geometric branches.

Source reference. Original lemma 0C28: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31935–31935. Cited at proof lines 32216-32217 for one direction of the equality; the assembly is inside the counted span.

Source reference. Original lemma 0C29: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31964–31964. Cited at proof lines 32217-32218 for the criterion; the verification in dimension one is inside the counted span.

Source reference. Original lemma 0C2B: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32100–32100. Cited at proof lines 32218-32219 to verify the equality criterion; the conclusion is inside the counted span.

Source reference. Original lemma 07NV: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10624–10624. Cited at proof lines 32220-32221 to supply the dimension hypothesis; the conclusion is inside the counted span.

Source reference. Original lemma 06LK: sources/stacks-more-algebra/pinned-original-source.tex, source lines 11132–11132. Cited at proof lines 32224-32225 to keep the strict henselization one-dimensional; the transfer is inside the counted span.

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Source reference. Original lemma 06LJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10979–10979. Cited at proof lines 32226-32227 for the completion identity behind the geometric-branch transfer; the conclusion is inside the counted span.

Lemma 0.2. *Let $(A, \mathfrak{m})$ be a Noetherian local ring of dimension 1. Then the number of (geometric) branches of A and $A^\wedge$ is the same.*

Proof. To see this for the number of branches, combine Lemmas 0C28, 0C29, and 0C2B and use that the dimension of $A^\wedge$ is one, see Lemma 07NV. To see this is true for the number of geometric branches we use the result for branches, the fact that the dimension does not change under strict henselization (Lemma 06LK), and the fact that $(A^{sh})^\wedge = ((A^\wedge)^{sh})^\wedge$ by Lemma 06LJ. □